Experiment-8

Develop a JavaScript program that will validate the controls inthe forms you have created for the application. State the assumptions you make (business logic you aretaking into consideration). Note: Your application must access a database using Servlet/JSP.

Aim:

To develop a JavaScript program for validating form inputs in a web application and ensure that user data satisfies the required business rules before submission. The validated data is then processed using Servlet/JSP and stored/retrieved from a database.

Algorithm:

1. Assumptions (Business Logic):

- Username must not be empty and should contain only alphabets.
- Password must be at least 6 characters long.
- Email must follow a valid email format.
- Required fields must not be left blank.
- Form submission should occur only if all validations are successful.

2. Input:

- User enters details such as username, password, email, etc., in the form.

3. Steps for Validation Using JavaScript:

Step 1: Create the Form (HTML)

- Design an HTML form with input fields (username, password, email, etc.).
- Attach a JavaScript validation function using the `onsubmit` event.

Step 2: Capture Form Data

- Use JavaScript (`document.forms` or `getElementById`) to retrieve input values.

Step 3: Perform Validation Checks

- Check if fields are empty.
- Validate username (only alphabets).
- Validate password length (minimum 6 characters).
- Validate email format using regular expressions

Step 4: Display Error Messages

- If validation fails, display appropriate error messages.
- Prevent form submission using `return false`.

Step 5: Allow Form Submission

- If all validations pass, allow the form to be submitted to a Servlet/JSP page.

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Step 6: Server-Side Processing (Servlet/JSP + Database)

- The servlet/JSP receives validated data.
- Perform additional validation (security).
- Connect to the database (e.g., MySQL).
- Store or retrieve user data as required.

CODE:

Student.jsp:

```
<%@ page language="java" contentType="text/html; charset=UTF-8" pageEncoding="UTF-8"%>

<html>
<head>
<title>Student Registration</title>

<style>
body {
    margin: 0;
    padding: 0;
    font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
    background: linear-gradient(135deg, #667eea, #764ba2);
    height: 100vh;
    display: flex;
    justify-content: center;
    align-items: center;
}

.container {
    background: #fff;
    padding: 30px 40px;
    border-radius: 12px;
    width: 350px;
    box-shadow: 0 10px 25px rgba(0,0,0,0.2);
}

h2 {
    text-align: center;
    margin-bottom: 20px;
    color: #333;
}

.input-group {
    margin-bottom: 15px;
}

.input-group label {
    display: block;
    margin-bottom: 5px;
    font-size: 14px;
    color: #555;
```

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```
}

.input-group input {
    width: 100%;
    padding: 10px;
    border: 1px solid #ccc;
    border-radius: 6px;
    transition: 0.3s;
    font-size: 14px;
}

.input-group input:focus {
    border-color: #667eea;
    outline: none;
    box-shadow: 0 0 5px rgba(102,126,234,0.5);
}

button {
    width: 100%;
    padding: 12px;
    border: none;
    background: linear-gradient(135deg, #667eea, #764ba2);
    color: #fff;
    font-size: 16px;
    border-radius: 6px;
    cursor: pointer;
    transition: 0.3s;
}

button:hover {
    opacity: 0.9;
    transform: scale(1.02);
}

</style>

<script src="js/validation.js"></script>

</head>

<body>

<div class="container">
    <h2> Student Registration</h2>

    <form action="StudentServlet" method="post" onsubmit="return validateForm()">

        <div class="input-group">
            <label>Name</label>
            <input type="text" id="name" name="name">
        </div>
```

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```
        <div class="input-group">
            <label>Age</label>
            <input type="text" id="age" name="age">
        </div>

        <div class="input-group">
            <label>Email</label>
            <input type="text" id="email" name="email">
        </div>

        <div class="input-group">
            <label>Phone</label>
            <input type="text" id="phone" name="phone">
        </div>

        <div class="input-group">
            <label>Class</label>
            <input type="text" id="studentClass" name="studentClass">
        </div>

        <div class="input-group">
            <label>Password</label>
            <input type="password" id="password" name="password">
        </div>

        <button type="submit">Register</button>

    </form>
</div>

</body>
</html>
```

StudentServlet.java:

```
package WTMA;



import java.io.IOException;

import java.sql.Connection;

import java.sql.DriverManager;

import java.sql.PreparedStatement;



import jakarta.servlet.ServletException;
```

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```
import jakarta.servlet.annotation.WebServlet;

import jakarta.servlet.http.HttpServlet;

import jakarta.servlet.http.HttpServletRequest;

import jakarta.servlet.http.HttpServletResponse;

@WebServlet("/StudentServlet")

public class StudentServlet extends HttpServlet {

    private static final long serialVersionUID = 1L;

    protected void doPost(HttpServletRequest request, HttpServletResponse response)

            throws ServletException, IOException {

        String name = request.getParameter("name");

        int age = Integer.parseInt(request.getParameter("age"));

        String email = request.getParameter("email");

        String phone = request.getParameter("phone");

        String studentClass = request.getParameter("studentClass");

        String password = request.getParameter("password");

        try {

            Class.forName("com.mysql.cj.jdbc.Driver");

            Connection con = DriverManager.getConnection(

                "jdbc:mysql://localhost:3306/school?allowPublicKeyRetrieval=true&useSSL=false&serverTimezone=UTC",

                "root",

                "root@1234"

            );

            String sql = "INSERT INTO students(name, age, email, phone, student_class, password) VALUES (?, ?, ?, ?, ?, ?)";

            PreparedStatement ps = con.prepareStatement(sql);

            ps.setString(1, name);

            ps.setInt(2, age);
```

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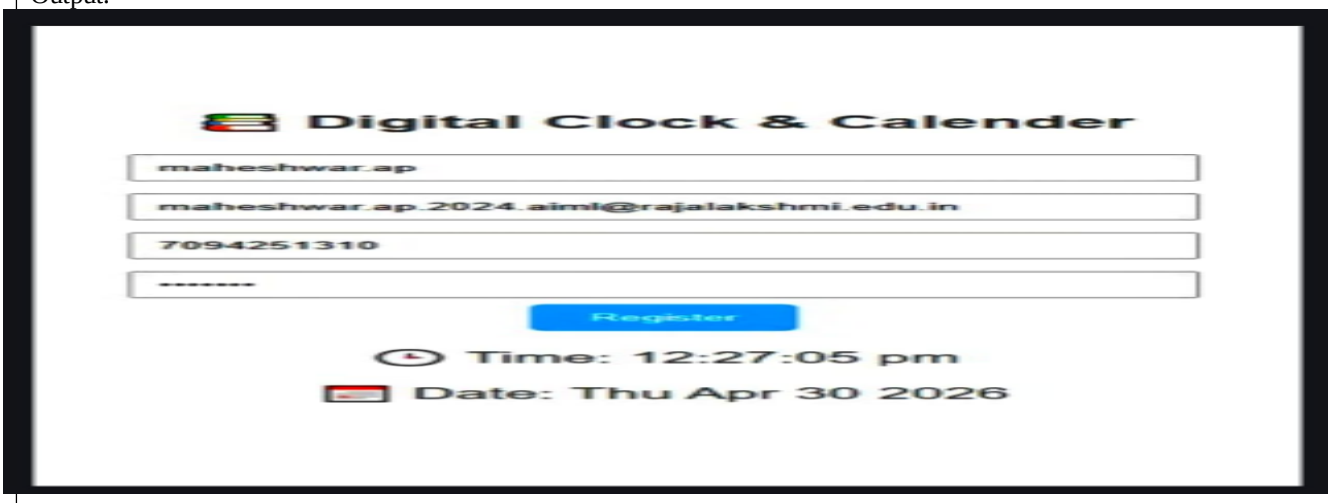
```
            ps.setString(3, email);

            ps.setString(4, phone);

            ps.setString(5, studentClass);

            ps.setString(6, password);

            int i = ps.executeUpdate();

            if (i > 0) {

                response.sendRedirect("success.jsp");

            } else {

                response.sendRedirect("error.jsp");

            }

            con.close();

        } catch (Exception e) {

            e.printStackTrace();

            response.getWriter().println("ERROR: " + e.toString());

        }

    }

}
```

Output:



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Result:

The JavaScript-based form validation was successfully implemented to verify user inputs based on defined business rules. Invalid inputs were prevented from being submitted, ensuring data integrity. Upon successful validation, the form data was processed by Servlet/JSP and stored/retrieved from the database, demonstrating effective client-side and server-side validation.

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